Unsupervised Clustering

Introduction

- Until now we have assumed that the training examples were labeled by their class membership.
- Procedures that use labeled samples are said to be *supervised*.
- In this chapter, we will study *clustering* as an *unsupervised* procedure that uses unlabeled samples.
- Unsupervised procedures are used for several reasons:
  - Collecting and labeling a large set of sample patterns can be costly.
  - One can train with large amount of unlabeled data, and then use supervision to label the groupings found.
  - Exploratory data analysis can provide insight into the nature or structure of the data.

Data Description

- Assume that we have a set of unlabeled multi-dimensional patterns.
- One way of describing this set of patterns is to compute their sample mean and covariance.
- This description uses the assumption that the patterns form a cloud that can be modeled with a hyperellipsoidal shape.
- However, we must be careful about any assumptions we make about the structure of the data.

Data Description

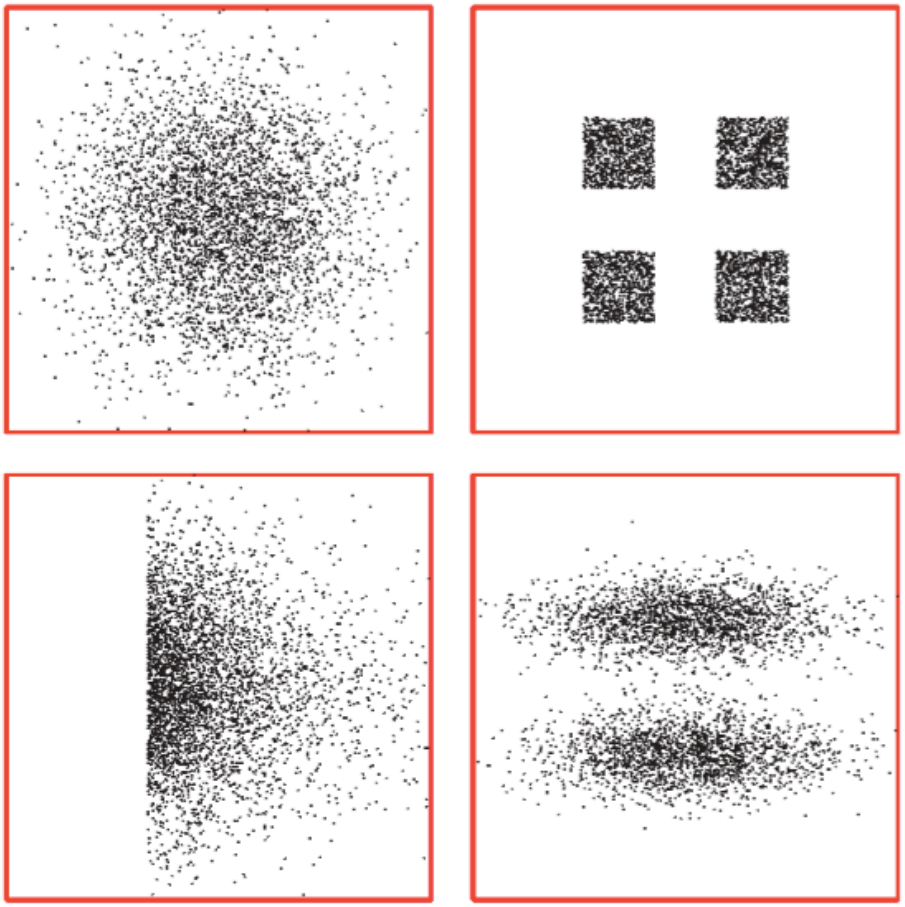


Figure 1: These four data sets have identical first-order and second-order statistics. We need to find other ways of modeling the structure. Clustering is an alternative way of describing the data in terms of groups of patterns.

Clusters

- A *cluster* is comprised of a number of similar objects collected or grouped together.
- Other definitions of clusters include (from Jain and Dubes, 1988):
  - A cluster is a set of entities which are alike, and entities from different clusters are not alike.
  - A cluster is an aggregation of points in the test space such that the distance between any two points in the cluster is less than the distance between any point in the cluster and any point not in it.

Clustering

- *Cluster analysis* organizes data by abstracting the underlying structure either as a grouping of individuals or as a hierarchy of groups.
- These groupings are based on measured or perceived similarities among the patterns.
- Clustering is unsupervised. Category labels and other information about the source of data influence the interpretation of the clusters, not their formation.

Clustering

- Clustering is a very difficult problem because data can reveal clusters with different shapes and sizes.



Figure 2: The number of clusters in the data often depend on the resolution (fine vs. coarse) with which we view the data. How many clusters do you see in this figure? 5, 8, 10, more?

Clustering

- Clustering algorithms can be divided into several groups:
  - *Exclusive* (each pattern belongs to only one cluster) vs. *nonexclusive* (each pattern can be assigned to several clusters)
  - *Hierarchical* (nested sequence of partitions) vs. *partitional* (a single partition)
- Implementations of clustering algorithms can also be grouped:
  - *Agglomerative* (merging atomic clusters into larger clusters) vs. *divisive* (subdividing large clusters into smaller ones)
  - *Serial* (processing patterns one by one) vs. *simultaneous* (processing all patterns at once)
  - *Graph-theoretic* (based on connectedness) vs. *algebraic* (based on error criteria)

Clustering

- Hundreds of clustering algorithms have been proposed in the literature.
- Most of these algorithms are based on the following two popular techniques:
  - Iterative squared-error partitioning
  - Agglomerative hierarchical clustering
- One of the main challenges is to select an appropriate measure of similarity to define clusters that is often both data (cluster shape) and context dependent.

Similarity Measures

- The most obvious measure of *similarity* (or dissimilarity) between two patterns is the distance between them.
- If distance is a good measure of dissimilarity, then we can expect the distance between patterns in the same cluster to be significantly less than the distance between patterns in different clusters.
- Then, a very simple way of doing clustering would be to choose a threshold on distance and group the patterns that are closer than this threshold.

Similarity Measures

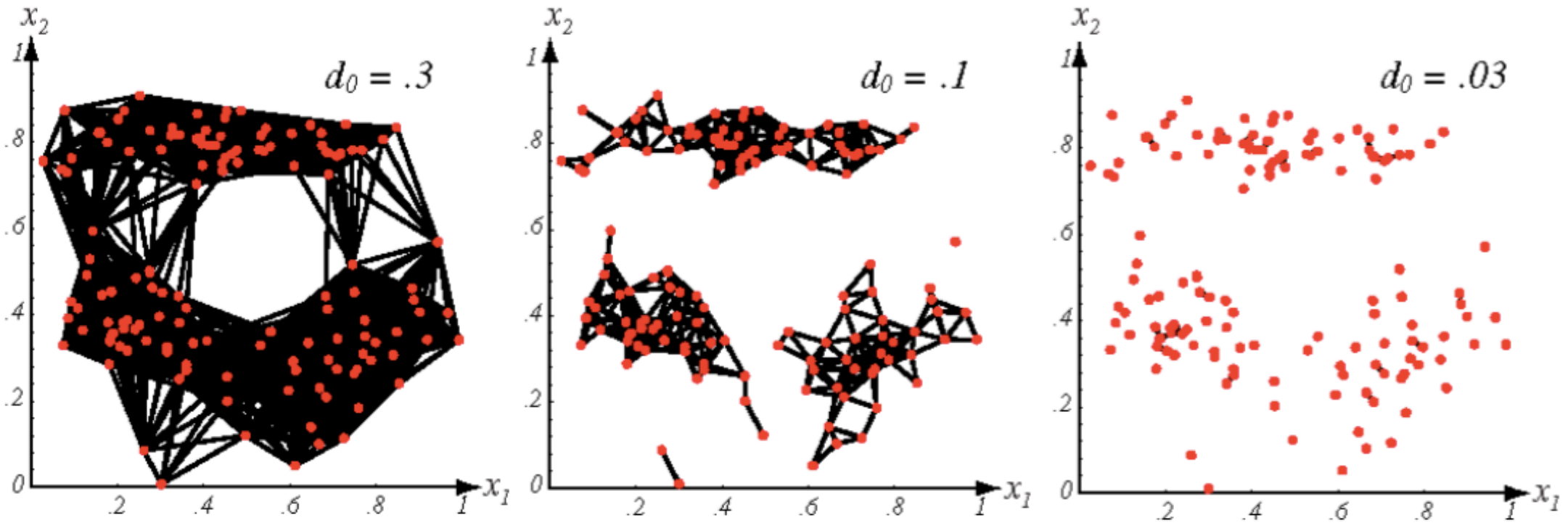


Figure 3: The distance threshold affects the number and size of clusters that are shown by lines drawn between points closer than the threshold.

Criterion Functions

- The next challenge after selecting the similarity measure is the choice of the criterion function to be optimized.
- Suppose that we have a set $\mathcal{D} = \{\mathbf{x_1}, \ldots, \mathbf{x}_n\}$ of n samples that we want to partition into exactly k disjoint subsets $\mathcal{D}_1, \ldots, \mathcal{D}_k$.
- Each subset is to represent a cluster, with samples in the same cluster being somehow more similar to each other than they are to samples in other clusters.
- The simplest and most widely used criterion function for clustering is the *sum-of-squared-error* criterion.

Squared-error Partitioning

- Suppose that the given set of n patterns has somehow been partitioned into k clusters $\mathcal{D}_1, \ldots, \mathcal{D}_k$.
- Let n_i be the number of samples in $\mathcal{D}_i$ and let $\mathbf{m_i}$ be the mean of those samples

$$\mathbf{m_i} = \frac{1}{n_i} \sum_{\mathbf{x} \in \mathcal{D}_i} \mathbf{x}$$

- Then, the sum-of-squared errors criterion to be minimized is:

$$J_e = \sum_{i=1}^{k} \sum_{\mathbf{x} \in \mathcal{D}_i} \|\mathbf{x} - \mathbf{m_i}\|^2$$

- For a given cluster $\mathcal{D}_i$, the mean vector $\mathbf{m_i}$ (centroid) is the best representative of the samples in $\mathcal{D}_i$.

K-Means Clustering

- A general algorithm for iterative squared-error partitioning to find k reference vectors (prototypes/codebook vectors/codewords) which best represent data:
  1. Initialize the k cluster centroids. Repeat steps 2 through 4 until the cluster membership stabilizes.
  2. Generate a new partition by assigning each pattern to its closest cluster center.
  3. Compute new cluster centers as the centroids of the clusters.
  4. Repeat steps 2 and 3 until an optimum value of the criterion function is found (e.g., when the error function does not change much or a predefined number of iterations are completed).

Squared-error Partitioning

- k-means is computationally efficient and gives good results if the clusters are compact, hyperspherical in shape and well-separated in the feature space.
- However, choosing k and choosing the initial partition are the main drawbacks of this algorithm.
- The value of k is often chosen empirically or by prior knowledge about the data.
- The initial partition is often chosen by generating k random points uniformly distributed within the range of the data, or by randomly selecting k points from the data.

Squared-error Partitioning

- Numerous attempts have been made to improve the performance of the basic k-means algorithm:
  - incorporating a fuzzy criterion resulting in fuzzy k-means,
  - using genetic algorithms, simulated annealing, deterministic annealing to optimize the resulting partition,
  - using iterative splitting to find the initial partition.
- Another alternative is to use model-based clustering using Gaussian mixtures to allow more flexible shapes for individual clusters (k-means with Euclidean distance assumes spherical shapes).
- In model-based clustering, the value of k corresponds to the number of components in the mixture.

Examples: K-means

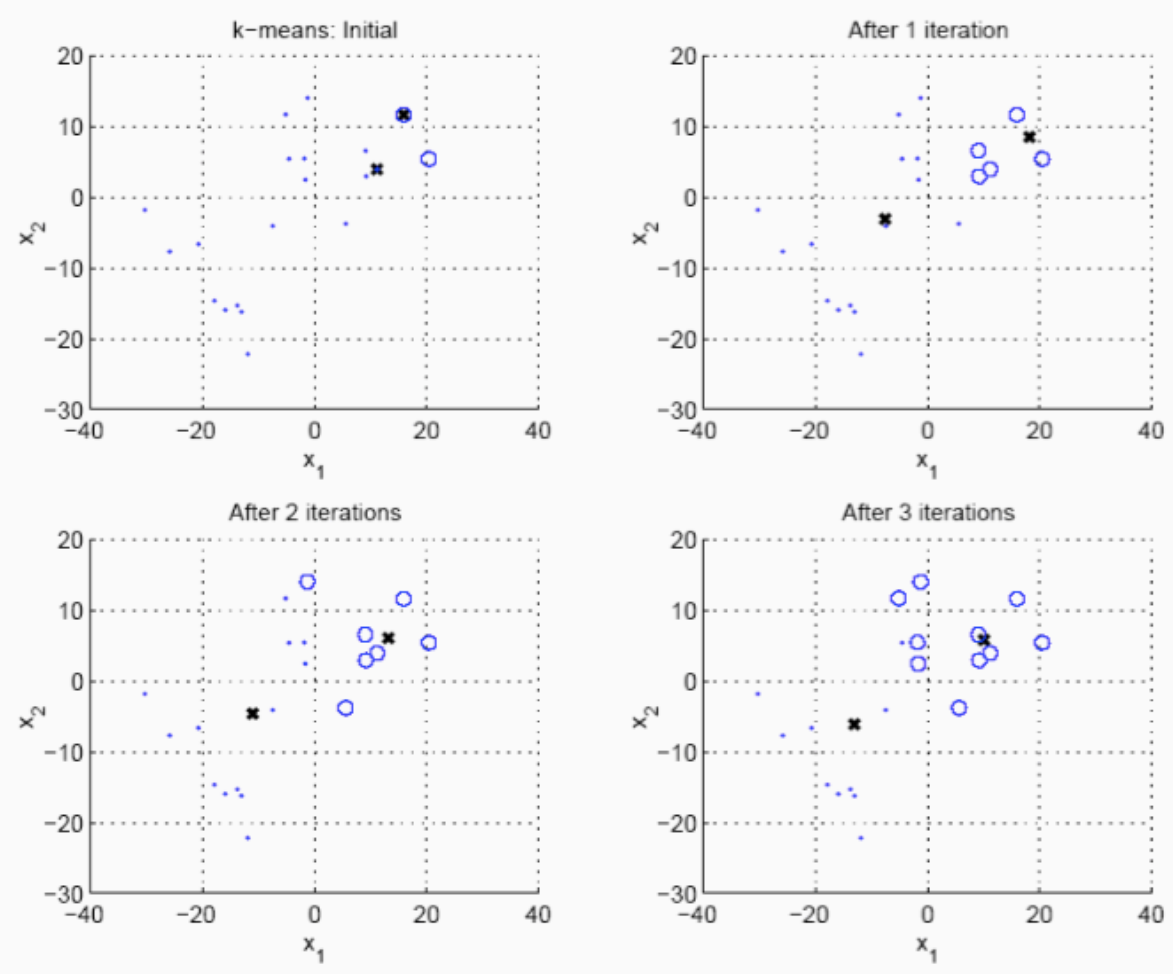
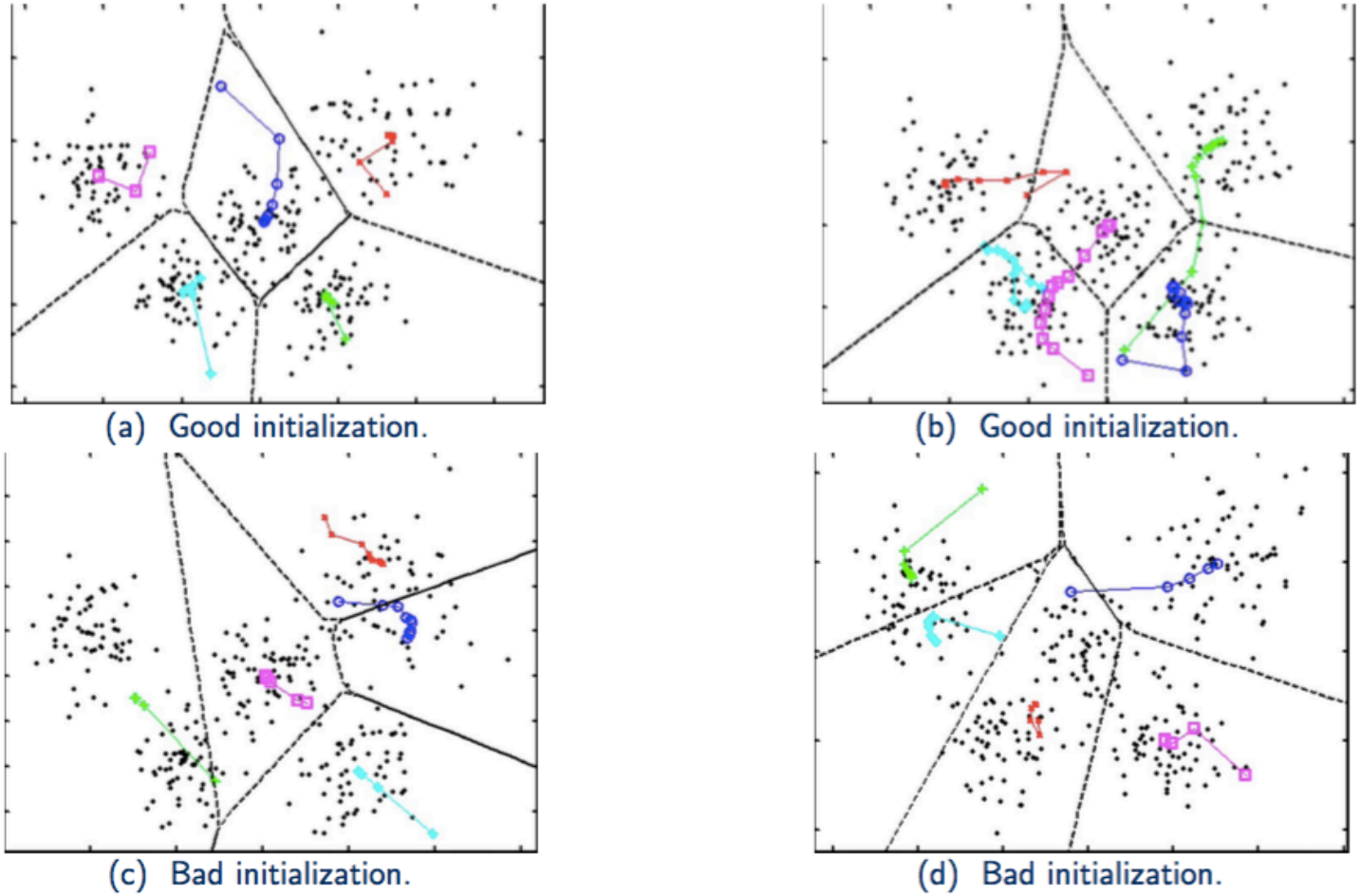


Figure 4: An example for the k-means algorithm.

Examples: K-Means



(a) Good initialization.

(b) Good initialization.

(c) Bad initialization.

(d) Bad initialization.

Figure 5: Examples for k-means with different initializations of five clusters for the same data.

How to initialize?

- Execute several runs of the K-means algorithm with different random initializations.
- Choose the clustering that gives the minimum that gives the minimum total sum of squared errors.
  - That is the «tightest» clustering.

Encoding/Decoding

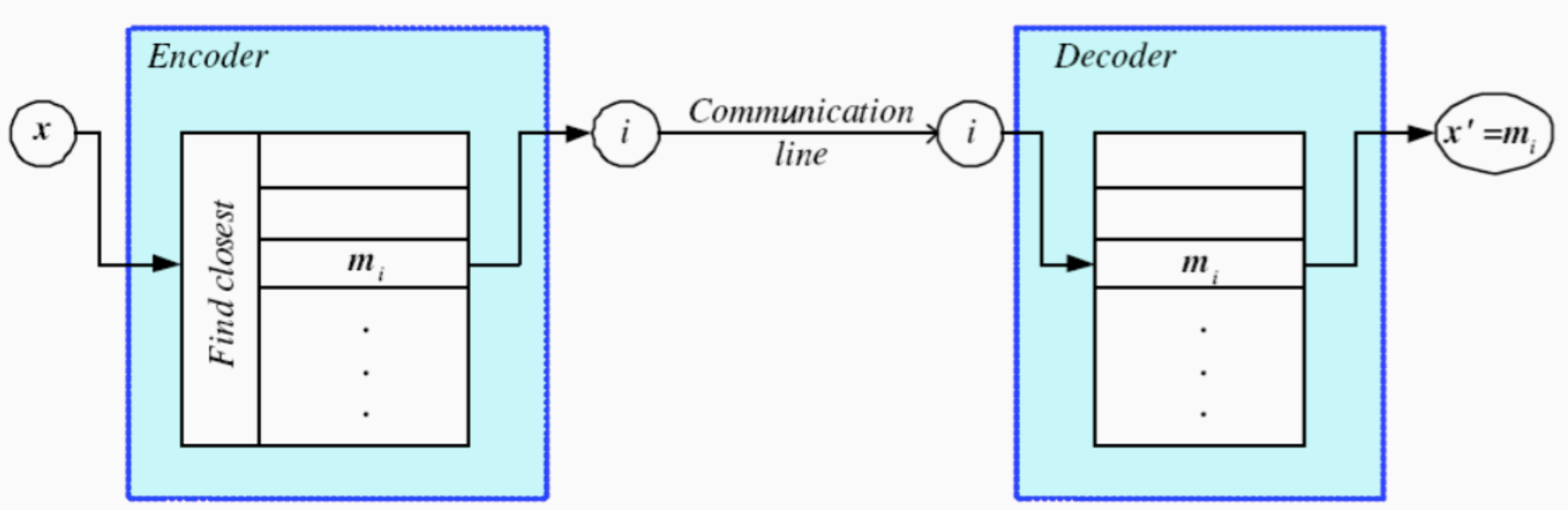


Figure 6: An example for vector quatization. The k cluster means represent our codebook.

Image Segmentation with K-means



Original Image



K=6



K = 3

Figure 7: Lena: for $k = 6$ the means of the six gray level classes are $[50, 80, 108, 138, 164, 202]$ for $k = 3$, the means of the three gray level classes are $[62, 122, 176]$

Color Image Segmentation with K-means



Original



K=8



K=3

Figure 8: Ballon girl. The cluster means for $k = 3$ are (in RGB color space):

$$m_1 = [72, 50, 67]$$

$$m_3 = [131, 125, 107]$$

$$m_3 = [185, 164, 167]$$

Fuzzy K-Means Clustering

1. Given N points $x_1, x_2, \ldots, x_N$. Initialize the centroids (means) of the k classes: $m_1, \ldots, m_k$. Set n $= 1$.

2. Given the k centroids, find the probability that the data point x_i belongs to class C_j, i.e. u_{ij}:

$$u_{ij} = \left(\frac{\frac{1}{\|x_i - m_j\|}}{\sum_{j=1}^{k} \frac{1}{\|x_i - m_j\|}} \right)^{1/(s-1)}, i = 1, \ldots, N; j = 1, \ldots, k,$$

where s is a constant around 2, and

$$0 \leq u_{ij} \leq 1; \qquad \sum_{j=1}^{N} u_{ij} = 1$$

Fuzzy K-Means Clustering

3. Find the centroid of each class to update $m_j, j = 1, \ldots, k$:

$$m_j = \frac{\sum_{i=1}^{N} u_{ij}^s x_i}{\sum_{i=1}^{N} u_{ij}^s}, j = 1, \ldots, k.$$

4. Repeat steps 2 and 3 until the change in the membership values between two successive iterations is below a certain threshold ε:

$$\|u_{ij}^n - u_{ij}^{n+1}\| < \varepsilon,$$

where n denotes the iteration number.

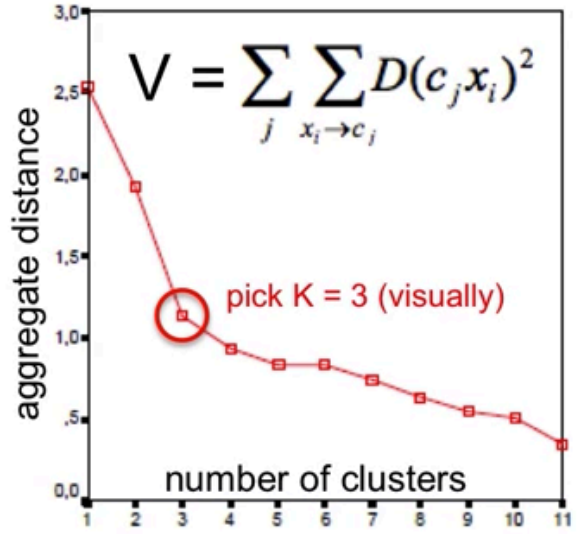
Color Image Segmentation with Fuzzy K-means

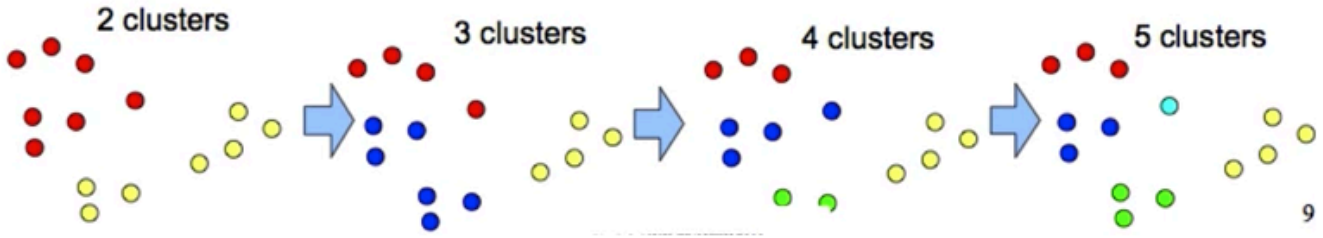


Figure 9: Ballon girl. Number of classes is $k = 8$. Image on the left is the original image. The middle image is result of the fuzzy k-means algorithm. The image on the right is the result of the k-means algorithm. The image segmented by the fuzzy k-means algorithm contains more details and is more pleasant to the eye.

Optimal number of clusters

- How many clusters are there in your data?
  - class labels may suggest the value of K (e.g. digits 0..9)
  - optimize distance V: for K = 2,3,…
    - run K-means, record distance
    - problem: V minimized when K = n
      - what if we use a validation set?
    - W&F: Minimum Description Length
      - total bits to encode K centroids + V
    - visually from scree plot:
      - point where "mountain" ends, "rubble" begins
      - elbow method: maximize 2nd derivative of V: point where rate of decline changes the most





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Adjusting K

- Can increase K incrementally.
- Leader cluster algorithm:
  - An instance that is far away from existing centers can cause creation of a new center.
  - Center that covers too many instances can be split into two.
  - Center that covers too few instances can be removed and restarted from some other part of the input space.

K-Means vs. Gaussian Mixture Models

- K-means is a special case of Gaussian Mixture Models obtained by Expectation Maximization.
  - Inputs are assumed independent with equal and shared variances,
  - All components have equal priors
  - Labels are hardened.
- K-means represent the density with circles.
- GMM uses ellipses of arbitrary shapes, orientations and proportions.

Note

- It is always a good idea to do dimensionality reduction before clustering using Euclidean distance if there are redundant or correlated features.

Hierarchical Clustering

- The k-means algorithm produces a *flat* data description where the clusters are disjoint and are at the same level.
- In some applications, groups of patterns share some characteristics when looked at a particular level.
- Hierarchical clustering tries to capture these multi-level groupings using *hierarchical* representations rather than flat partitions.

Hierarchical Clustering

- In hierarchical clustering, for a set of n samples,
  - the first level consists of n clusters (each cluster containing exactly one sample),
  - the second level contains $n - 1$ clusters,
  - the third level contains $n - 2$ clusters,
  - and so on until the last (n'th) level at which all samples form a single cluster.
- Given any two samples, at some level they will be grouped together in the same cluster and remain together at all higher levels.

Hierarchical Clustering

- The most natural representation of hierarchical clustering is a tree, also called a *dendrogram*, which shows how the samples are grouped.
- If there is an unusually large gap between the similarity values for two particular levels, one can argue that the level with fewer number of clusters represents a more natural grouping.

Hierarchical Clustering

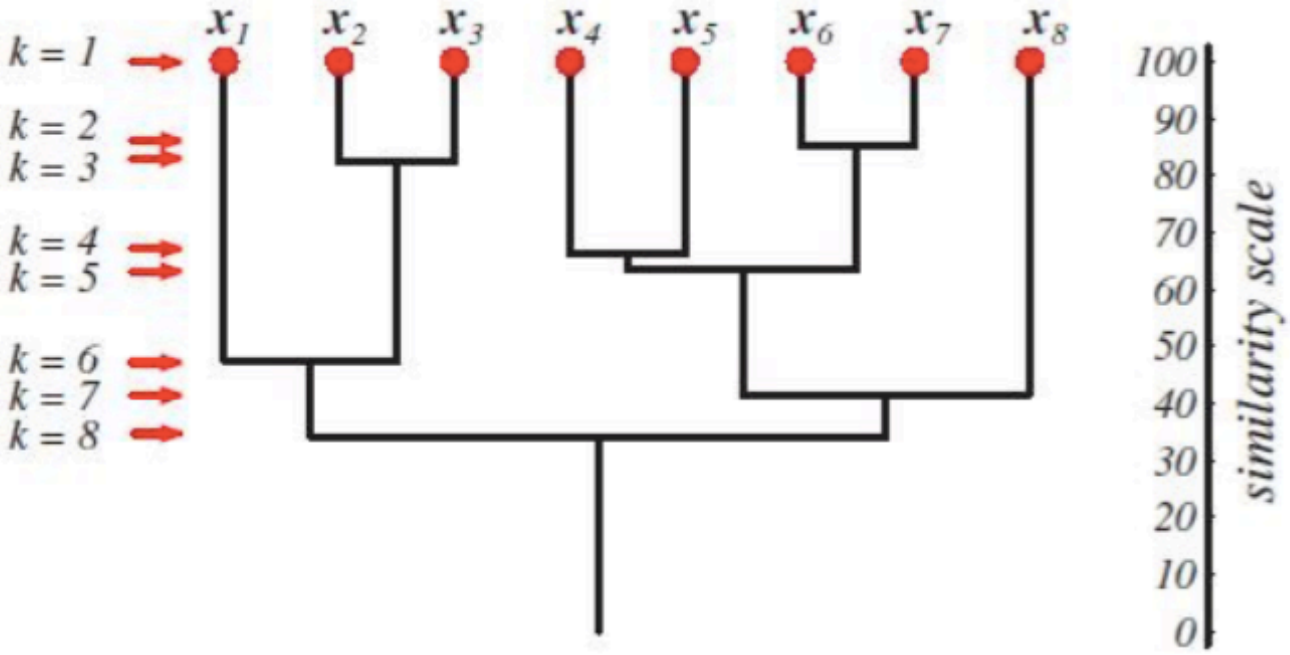
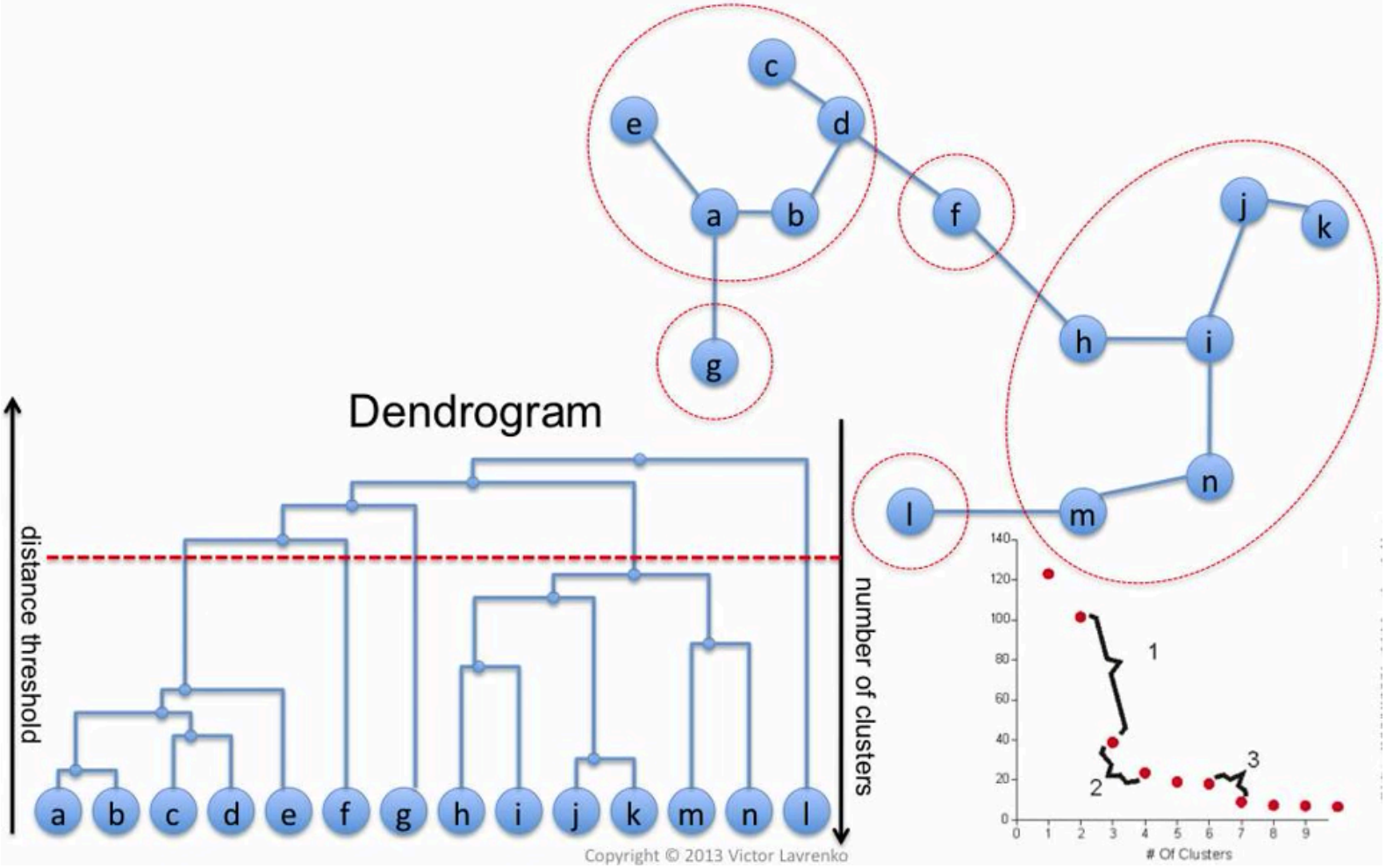


FIGURE 10.11. A dendrogram can represent the results of hierarchical clustering algorithms. The vertical axis shows a generalized measure of similarity among clusters. Here, at level 1 all eight points lie in singleton clusters; each point in a cluster is highly similar to itself, of course. Points $\mathbf{x}_6$ and $\mathbf{x}_7$ happen to be the most similar, and are merged at level 2, and so forth. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

Hierarchical Clustering

- *Agglomerative Hierarchical Clustering:*
  1. Specify the number of clusters. Place every pattern in a unique cluster and repeat steps 2 and 3 until a partition with the required number of clusters is obtained.
  2. Find the closest clusters according to a distance measure.
  3. Merge these two clusters.
  4. Return the resulting clusters.

Agglomerative clustering: example



Hierarchical Clustering

- Popular distance measures to find the nearest clusters (for two clusters $\mathcal{D}_i$ and $\mathcal{D}_j$):

$$d_{\text{min}}(\mathcal{D}_i, \mathcal{D}_j) = \min_{\substack{\mathbf{x} \in \mathcal{D}_i \\ \mathbf{x}' \in \mathcal{D}_j}} \|\mathbf{x} - \mathbf{x}'\|$$

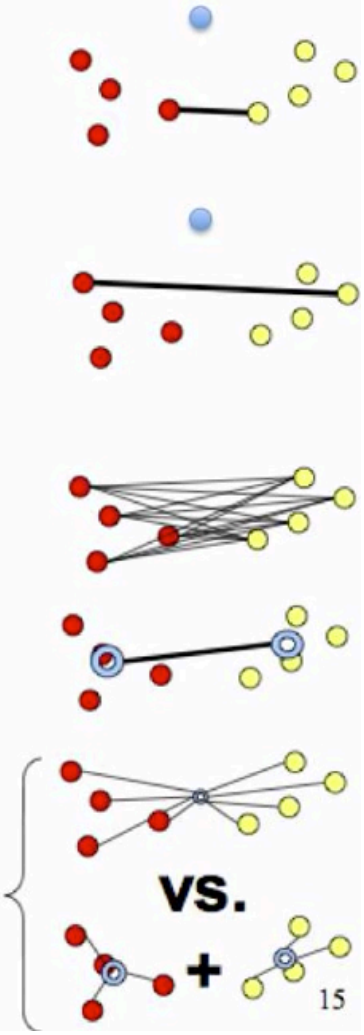
$$d_{\text{max}}(\mathcal{D}_i, \mathcal{D}_j) = \max_{\substack{\mathbf{x} \in \mathcal{D}_i \\ \mathbf{x}' \in \mathcal{D}_j}} \|\mathbf{x} - \mathbf{x}'\|$$

$$d_{\text{avg}}(\mathcal{D}_i, \mathcal{D}_j) = \frac{1}{\#\mathcal{D}_i \, \#\mathcal{D}_j} \sum_{\mathbf{x} \in \mathcal{D}_i} \sum_{\mathbf{x}' \in \mathcal{D}_j} \|\mathbf{x} - \mathbf{x}'\|$$

$$d_{\text{mean}}(\mathcal{D}_i, \mathcal{D}_j) = \|\mathbf{m}_i - \mathbf{m}_j\|$$

Cluster distance measures

- Single link: $D(c_1,c_2) = \min_{x_1 \in c_1, x_2 \in c_2} D(x_1,x_2)$
  - distance between closest elements in clusters
  - produces long chains a→b→c→…→z
- Complete link: $D(c_1,c_2) = \max_{x_1 \in c_1, x_2 \in c_2} D(x_1,x_2)$
  - distance between farthest elements in clusters
  - forces "spherical" clusters with consistent "diameter"
- Average link: $D(c_1,c_2) = \frac{1}{|c_1|}\frac{1}{|c_2|}\sum_{x_1 \in c_1}\sum_{x_2 \in c_2} D(x_1,x_2)$
  - average of all pairwise distances
  - less affected by outliers
- Centroids: $D(c_1,c_2) = D\left(\left(\frac{1}{|c_1|}\sum_{x \in c_1}\vec{x}\right),\left(\frac{1}{|c_2|}\sum_{x \in c_2}\vec{x}\right)\right)$
  - distance between centroids (means) of two clusters
- Ward's method: $TD_{c_1 \cup c_2} = \sum_{x \in c_1 \cup c_2} D(x,\mu_{c_1 \cup c_2})^2$
  - consider joining two clusters, how does it change the total distance (TD) from centroids?



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Hierarchical Clustering

- When d_{min} is used to measure the distance between clusters, the algorithm is called the nearest neighbor clustering algorithm.
- Moreover, if the algorithm is terminated when the distance between nearest clusters exceeds a threshold, it is called the *single linkage algorithm* where
  - patterns represent the nodes of a graph,
  - edges connect patterns belonging to the same cluster,
  - merging two clusters corresponds to adding an edge between the nearest pair of nodes in these clusters.

Single Link Clustering

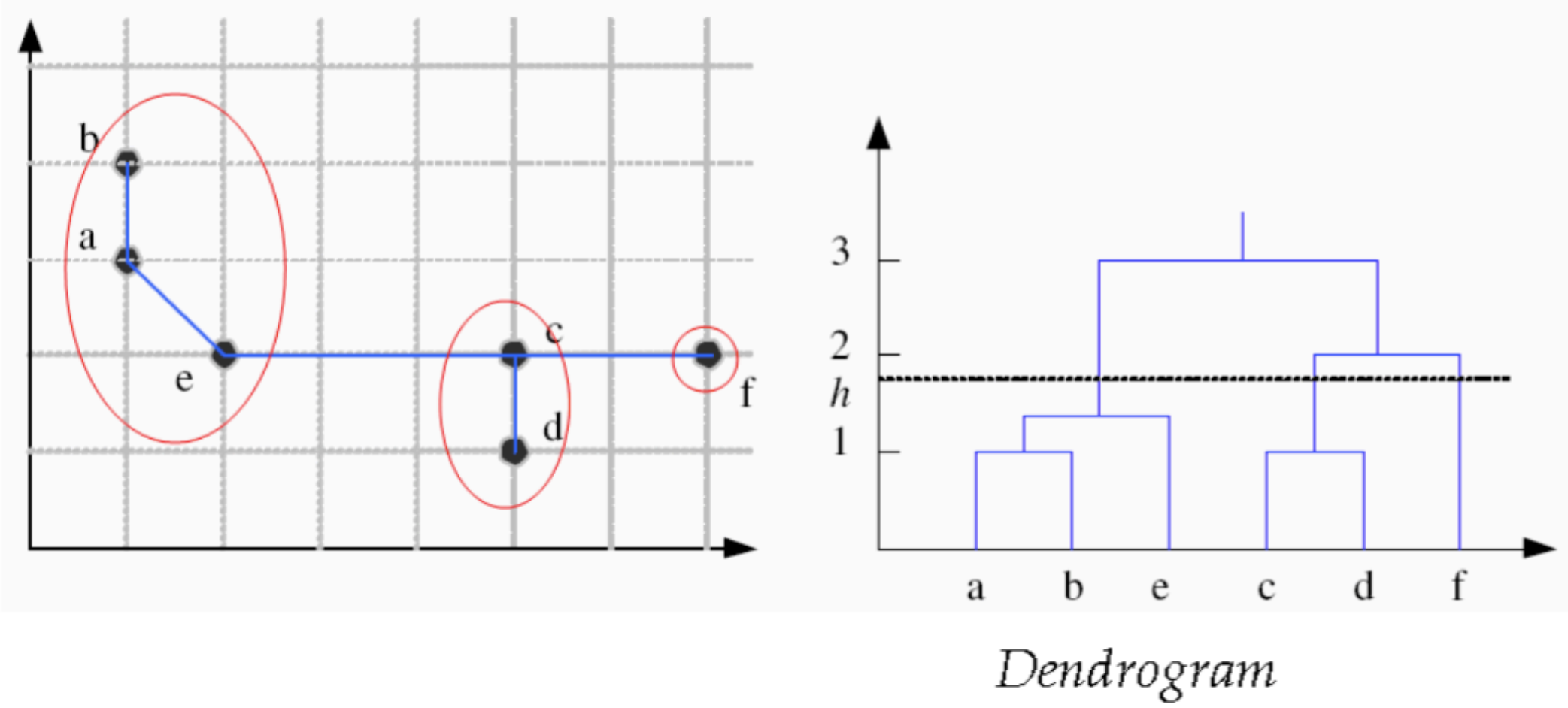


Figure 10: Example for agglomerative hierarchical clustering.

Hierarchical Clustering

- *Stepwise-Optimal Hierarchical Clustering:*
  1. Specify the number of clusters. Place every pattern in a unique cluster and repeat steps 2 and 3 until a partition with the required number of clusters is obtained.
  2. Find the clusters whose merger increases an error criterion the least.
  3. Merge these two clusters.
  4. Return the resulting clusters.
- When the sum-of-squared-error criterion J_e is used, the pair of clusters whose merger increases J_e as little as possible is the pair for which the distance

$$d_e(\mathcal{D}_i, \mathcal{D}_j) = \frac{\#\mathcal{D}_i \#\mathcal{D}_j}{\#\mathcal{D}_i + \#\mathcal{D}_j} \|\mathbf{m}_i - \mathbf{m}_j\|^2$$

  is minimum.

Cluster Validity

- The procedures we have considered so far either assume that the number of clusters is known or use some thresholds to decide how the clusters are formed.
- These may be reasonable assumptions for some applications but are usually unjustified if we are exploring a data set whose properties and structure are unknown.
- Furthermore, most of the iterative algorithms that we use may find a local extremum and may not give the best result.

Cluster Validity

- Methods for validating the results of a clustering algorithm include:
  - Repeating the clustering procedure for different values of the parameters, and examining the resulting values of the criterion function for large jumps or stable ranges.
  - Evaluating the goodness-of-fit using measures such as the chi-squared or Kolmogorov-Smirnov statistics.
  - Formulating hypothesis tests that check whether multiple clusters found have been formed by chance, and whether the observed change in the error criterion has any significance.

References

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- Pattern Recognition Course Web Page: http://www.cs.bilkent.edu.tr/~saksoy/courses/cs551/
- Machine Learning course by Andrew Ng. https://www.coursera.org/learn/machine-learning
- *An Introduction to Statistical Learning, with applications in R* (Springer, 2013), G. James, D. Witten, T. Hastie and R. Tibshirani